Data Analysis and Tax Compliance

Block 2B: Bivariate analysis

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Today: identify, compare, interpret

Open the interactive activity

- Choose a method from the variables and the research question.
- Read contingency tables; calculate χ^2 and Cramér's V.
- Compare group means with t/ANOVA and describe effect size.
- Interpret covariance, correlation and changes after stratification.

Discussion

6 November, 10:30–12:30. No R is required; coding follows in a separate lesson.

Three pairs of variable types

Types	Tax example	Starting point
Two categorical	Channel and filing outcome	Table, conditional shares, χ^2, V
Categorical + quantitative	Procedure and processing days	Group distributions, t/ANOVA, η^2
Two quantitative	Document count and review time	Scatterplot, covariance, r

The unit is a filing/case. Examples use separate synthetic datasets. Never interpret arbitrary category codes as measured quantities.

A table before a test

Channel	On time	Late	Not received	Total
Email	150	70	40	260
SMS	70	90	50	210
Letter	40	30	60	130
Total	260	190	150	600

Outcome is measured at a fixed date; “not received” is an observed category, not an empty data field.

Discussion

Which denominator compares on-time rates across channels?

Joint, marginal and conditional proportions

- Joint share: $P(\text{Email and On time})$ estimated by $150/600 = .25$.
- Marginal share: Email cases $260/600 = .4333$.
- Conditional share: On time **given Email**, $150/260 = .5769$.

For Email and Late:

$$P(\text{Late} \mid \text{Email}) \approx 70/260 = .2692,$$

$$P(\text{Email} \mid \text{Late}) \approx 70/190 = .3684.$$

Discussion

E-tivity 4, section 2: switch between row and column percentages. Say each denominator aloud.

Independence: the product rule

Under population independence,

$$P(A_i \cap B_j) = P(A_i)P(B_j).$$

Estimate the expected cell count under the null using the sample margins:

$$E_{ij} = \frac{n_{i\cdot} n_{\cdot j}}{n}.$$

Email and On time

$E = 260 \times 260/600 = 112.6667$, compared with $O = 150$.

The test concerns population independence under a sampling model, not exact equality of every sample count to its expectation.

Build Pearson's χ^2

$$\chi^2 = \sum_i \sum_j \frac{(O_{ij} - E_{ij})^2}{E_{ij}}.$$

Email and On time contributes

$$\frac{(150 - 112.6667)^2}{112.6667} = 12.3708.$$

- Signed $O - E$ differences cancel across a row or column.
- Squaring prevents cancellation; dividing by E scales the discrepancy.
- Sum all nine contributions, not just the largest cell.

From χ^2 to a conclusion

$$\chi^2 = 68.1374, \quad df = (3-1)(3-1) = 4, \quad p \approx 5.61 \times 10^{-14}.$$

- Reject independence at 5%, if the design and approximation are appropriate.
- Independent observations, mutually exclusive categories, adequate expected counts are required.
- Here the smallest expected count is 32.5.

Discussion

Can this test establish that email caused the higher on-time rate? Why not?

Look at the pattern, not only the p-value

On-time proportions: Email 57.69%; SMS 33.33%; Letter 30.77%.

$$\text{Pearson residual}_{ij} = \frac{O_{ij} - E_{ij}}{\sqrt{E_{ij}}}.$$

Positive residual: more cases than expected under independence. Negative residual: fewer.

Scope

Residuals help describe cells driving the overall statistic. They are not automatically adjusted tests for many cell comparisons.

Cramér's V measures relative association

$$V = \sqrt{\frac{\chi^2}{n \min(r-1, c-1)}} = \sqrt{\frac{68.1374}{600 \times 2}} = .2383.$$

- $0 \leq V \leq 1$; V has no sign.
- The expression without the square root is V^2, not V.
- The same V can describe different cell patterns.

Discussion

A hypothetical independent sample has five times the counts but the same percentages. Predict what happens to χ^2, p and V.

Sample size changes evidence

If every count is multiplied by k, with proportions fixed:

$$\chi^2_{\text{new}} = k\chi^2, \qquad V_{\text{new}} = V.$$

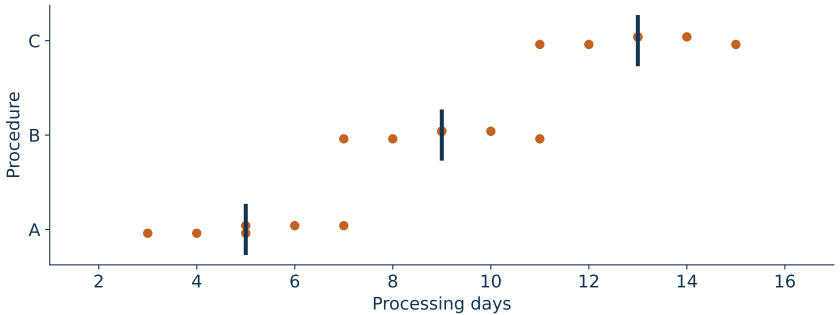
A larger independent sample can make the same association more statistically evident.

Do not create false information

Copying existing rows does not produce new independent observations. The e-tivity multiplier represents hypothetical larger independent samples.

Effect size and statistical significance answer different questions.

Quantitative outcomes across groups



Constructed processing times: means 5, 9 and 13 days; six cases per group; each sample variance is 2 days^2. Vertical lines mark means.

Ask about separation and spread

E-tivity 4, section 5: 5 minutes

Increase the gap between means while holding within-group spread fixed. Then increase within-group spread while holding means fixed.

- Large mean differences relative to internal spread are easier to distinguish.
- A grand mean must weight groups by their sizes.
- A distribution plot can expose skewness, outliers and heterogeneity hidden by a mean.

Two groups: independent or paired?

- Independent procedures: use a two-sample comparison of means.
- Welch's t test estimates group variances separately.
- The pooled t test assumes a common variance.
- Before/after on the same cases: analyse paired differences.

Exact link under the matching classical model

With two independent groups and a common variance, one-way ANOVA has $F = t^2$ and the same p-value as the two-sided pooled t test.

ANOVA separates two sources of variation

$$SS_T = \sum_j \sum_i (x_{ij} - \bar{x})^2 = SS_B + SS_W,$$

$$SS_B = \sum_j n_j (\bar{x}_j - \bar{x})^2, \qquad SS_W = \sum_j \sum_i (x_{ij} - \bar{x}_j)^2.$$

- Between: weighted separation of group means from the grand mean.
- Within: deviations of observations from their own group means.

For our groups: $SS_B = 192$, $SS_W = 30$, $SS_T = 222$.

The ANOVA table

Source	SS	df	MS
Between groups	192	$k-1=2$	96
Within groups	30	$n-k=15$	2
Total	222	$n-1=17$	—

$$F = \frac{MS_B}{MS_W} = \frac{96}{2} = 48.$$

Under the classical null model, compare this with $F_{2,15}$; p-value is the right-tail area.

What does the global F test establish?

$$H_0 : \mu_A = \mu_B = \mu_C, \qquad H_1 : \text{not all means are equal.}$$

Here $F(2, 15) = 48$, $p \approx 3.03 \times 10^{-7}$.

- Reject equal population means under the assumptions.
- This does not tell us that every pair differs.
- It does not establish that changing procedure caused the difference.

Discussion

Which additional comparisons or design information would you need?

Effect size: eta-squared

$$\eta^2 = \frac{SS_B}{SS_T} = \frac{192}{222} = .8649.$$

About 86.5% of observed total squared variation is associated with group membership in this deliberately separated example.

- This is a descriptive variance proportion.
- It is not the probability that the test is correct.
- It is not a causal fraction of processing time.

Discussion

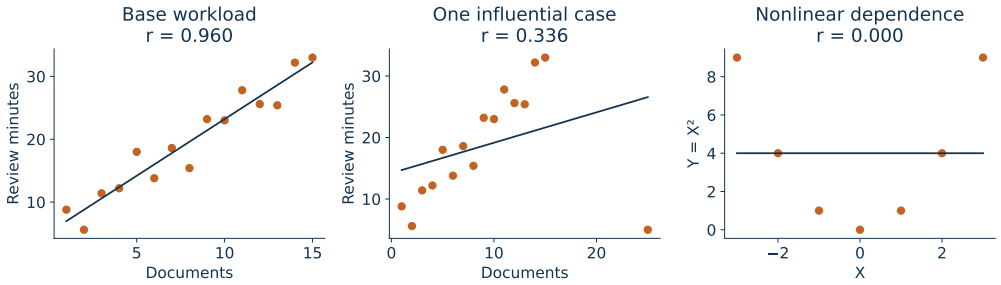
Can a tiny between-group effect become significant in a sufficiently large sample?

Check the model and plan the follow-up

- Classical one-way ANOVA: independent observations, normal within-group errors and common population error variance.
- Use group and residual plots; independence depends on the study design.
- Unequal variances: consider Welch ANOVA.
- Pairwise follow-up: use justified contrasts or multiplicity-adjusted methods, such as Tukey under the classical model.

Testing many pairs without adjustment inflates the chance of false positives.

Two quantitative variables: start with a plot



A fitted line and r can change after one influential observation. Zero correlation can coexist with a deterministic nonlinear relationship.

Covariance: matching and opposing deviations

$$s_{XY} = \frac{1}{n-1} \sum_i (x_i - \bar{x})(y_i - \bar{y}).$$

- Both deviations positive or both negative: positive contribution.
- Opposite signs: negative contribution.
- Covariance units are the product of the two variables' units.

Discussion

Would expressing review time in seconds rather than minutes change covariance?

Pearson correlation standardises covariance

$$r = \frac{s_{XY}}{s_X s_Y} = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_i (x_i - \bar{x})^2 \sum_i (y_i - \bar{y})^2}}.$$

- $-1 \leq r \leq 1$, provided both variables vary.
- r describes the direction and strength of a **linear** association.
- It is symmetric in X and Y; it is not a slope.
- Population correlation ρ and its sample estimate r are distinct.

The same r can hide different relationships

Relationship	r	Slope
$Y = 2X$	1	2
$Y = 3X$	1	3
$Y = X^2$, symmetric $X = -3, \ldots, 3$	0	0 (linear fit)

- Strength of a linear relationship is distinct from its size in units.
- Independence implies zero covariance when the relevant moments exist.
- Zero covariance/correlation does not generally imply independence.

Simpson: a headline worth checking

A misleading causal headline

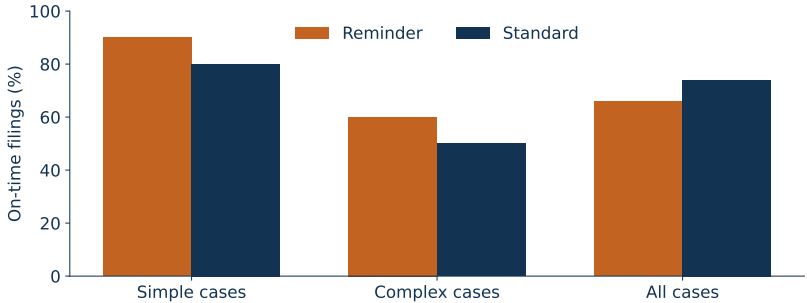
"Reminders reduce compliance: 66% versus 74% on time."

Case complexity	Reminder	Standard
Simple	$18/20 = 90\%$	$64/80 = 80\%$
Complex	$48/80 = 60\%$	$10/20 = 50\%$
Total	$66/100 = 66\%$	$74/100 = 74\%$

Discussion

Both conditional comparisons favour the reminder group. How can the total reverse?

Different weights explain the reversal



Reminder: $.2(.9) + .8(.6) = .66$. Standard: $.8(.8) + .2(.5) = .74$. Reminder cases are much more concentrated in the complex stratum.

Stratification does not automatically establish causation

- Marginal and conditional associations can answer different questions.
- Choose adjustment variables from substantive reasoning and the causal question.
- Neither the aggregate table nor a reversal proves a policy effect.
- Selection, timing, measurement and unobserved differences may remain.

Discussion

Use the mixture sliders. Then rewrite the headline without asserting causation.

A complete bivariate comment

- State the units, variables, question and target population.
- Describe the table or plot and report the effect in meaningful terms.
- Name the test and assumptions; interpret the p-value.
- Report an effect size and a limitation on generalisation or causation.

Discussion

Complete the 200–250 word e-tivity report and export your responses. Bring one question to the separate R lesson.

Core lecture ends here. The remaining slides are optional reference/learning bridges.

Optional bridge: a fitted line has a direction

$$\hat{y} = b_0 + b_1 x, \qquad b_1 = \frac{s_{XY}}{s_X^2}, \quad b_0 = \bar{y} - b_1 \bar{x}.$$

OLS minimises $SSE = \sum_i (y_i - \hat{y}_i)^2$, the squared vertical residuals. For the base workload sample:

$$\widehat{\text{minutes}} = 5.1714 + 1.8036 \times \text{documents}.$$

An additional document is associated with about 1.804 more predicted minutes. This does not establish a causal effect.

Optional bridge: fit, error and prediction

For simple OLS with an intercept,

$$R^2 = 1 - \frac{SSE}{SST} = r^2 = .9207, \qquad SER = \sqrt{\frac{SSE}{n-2}} = 2.4557 \text{ minutes}.$$

- R^2 is a relative in-sample fit measure.
- SER describes residual spread in the units of Y.
- Neither is a substitute for evaluating prediction on new cases.

We will return to training/test separation in supervised learning.

Optional bridge: assumptions before regression inference

- The conditional mean $E(Y \mid X)$ follows the specified linear form.
- Errors are independent, with mean zero conditional on X.
- Classical SEs assume constant conditional error variance.
- Normal errors give exact small-sample t/F inference; X need not be marginally normal.

Inspect residual patterns and influential points. A large R^2 does not validate these conditions.

Optional bridge: slope uncertainty

$$SE(b_1) = \frac{SER}{\sqrt{\sum_i (x_i - \bar{x})^2}} = .1468.$$

With $n = 15$, use $t_{.975,13} = 2.1604$ for a 95% interval:

$$1.8036 \pm 2.1604(.1468) \approx [1.4865, 2.1206] \text{ min/document.}$$

For $H_0 : \beta_1 = 0$, $t = b_1/SE(b_1)$, with 13 df. The p-value is about 1.57×10^{-8} under the classical model. The simple-regression global F statistic equals this slope-test t^2.

Optional bridge: mean versus new-case intervals

For a new document count x_0, define

$$h_0 = \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{\sum_i (x_i - \bar{x})^2}.$$

$$\text{Mean-response CI}: \quad \widehat{y}_0 \pm t\, SER \sqrt{h_0},$$

$$\text{New-case prediction interval}: \quad \widehat{y}_0 \pm t\, SER \sqrt{1 + h_0}.$$

The extra 1 represents individual variation. Both depend on the model; extrapolation adds unvalidated assumptions. Try $x_0 = 8$ and 20; observed X ranges from 1 to 15.

Reference: dependence is more than correlation

For suitable finite moments:

$$X \perp Y \;\Rightarrow\; E(Y \mid X) = E(Y) \;\Rightarrow\; \mathrm{Cov}(X, Y) = 0.$$

- Mean independence as written is directional.
- The reverse implications do not hold in general.
- If the joint distribution is bivariate normal, zero correlation does imply independence.

For i.i.d. bivariate normal data, the test of $H_0 : \rho = 0$ uses $t = r\sqrt{(n-2)/(1-r^2)}$ with $n - 2$ df.

R laboratory, Part B: bivariate analysis

The supplied BES education materials provide examples of:

- Contingency tables: area and a dropout indicator.
- Two groups: NEET percentages by dropout category.
- Several groups: NEET percentages across geographical areas.
- Quantitative pairs: tertiary education and cultural participation.

Regional percentages describe areas, not individual behaviour. Model-based tests require explicit assumptions when the dataset covers all regions.

From bivariate analysis to supervised learning

- A simple fitted line links the bivariate examples to prediction.
- Multiple regression belongs to the supervised block, with validation and diagnostics.
- A separate R session follows the theory of supervised learning.
- The unsupervised block also pairs theory with examples in R.

Discussion

Which question concerns association in the observed data, and which concerns prediction for new cases?